Experiment 9:

📁 Step 1: Create Project Folder

```
mkdir docker-project
cd docker-project
```

📝 Step 2: Create app.py

```
notepad app.py
```

👉 Code:

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def home():
    return "Hello from Docker App!"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

📦 Step 3: Create requirements.txt

```
notepad requirements.txt
```

👉 Add:

```
flask
```

📄 Step 4: Create Dockerfile (IMPORTANT)

```
notepad Dockerfile
```

👉 Paste this:

```
# Use Python image
FROM python:3.9

# Set working directory
```

```
WORKDIR /app

# Copy files
COPY . .

# Install dependencies
RUN pip install -r requirements.txt

# Run application
CMD ["python", "app.py"]
```

🐳 Step 5: Build Docker Image

```
docker build -t myapp .
```

👉 Explanation:

- `-t myapp` → image name
- `.` → current folder

▶□ Step 6: Run Docker Container

```
docker run -p 5000:5000 myapp
```

👉 Explanation:

- `-p 5000:5000` → port mapping

🌐 Step 7: Check Output

👉 Open browser:

```
http://localhost:5000
```

👉 Output:

```
Hello from Docker App!
```

□ Step 8: Verify Container Running

```
docker ps
```

☞ Shows running containers

⏺ Step 9: Stop Container

☞ Press:

```
Ctrl + C
```

EXPERIMENT NO: 10

📁 Step 1: Create Project Folder

```
mkdir compose-project
cd compose-project
```

✍ Step 2: Create app.py

```
notepad app.py
```

☞ Code:

```
from flask import Flask
import mysql.connector

app = Flask(__name__)

@app.route('/')
def home():
    return "Hello from Docker Compose App!"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

🎁 Step 3: Create requirements.txt

```
notepad requirements.txt
```

👉 Add:

```
flask
mysql-connector-python
```

📄 Step 4: Create Dockerfile

```
notepad Dockerfile
```

👉 Code:

```
FROM python:3.9

WORKDIR /app

COPY . .

RUN pip install -r requirements.txt

CMD ["python", "app.py"]
```

⚙️ Step 5: Create docker-compose.yml (MOST IMPORTANT)

```
notepad docker-compose.yml
```

👉 Paste this:

```
version: '3'

services:
  web:
    build: .
    ports:
      - "5000:5000"
    depends_on:
      - db
```

```
db:
  image: mysql:5.7
  environment:
    MYSQL_ROOT_PASSWORD: root
    MYSQL_DATABASE: testdb
  ports:
    - "3306:3306"
```

▶ Step 6: Run Docker Compose

```
docker-compose up
```

☞ OR (new version):

```
docker compose up
```

🌐 Step 7: Check Output

☞ Open browser:

```
http://localhost:5000
```

☞ Output:

```
Hello from Docker Compose App!
```

 Step 8: Verify Running Containers

```
docker ps
```

☞ You will see:

- web container
- db container

🛑 Step 9: Stop Containers

👉 Press:

```
Ctrl + C
```